Health-Project 3: Breast Cancer Detection

Experiment 3: Data Pre-processing

Course: CSE4001 — Applications of Machine Learning in Industry | **Target:** Malignant (1) vs. Benign (0)

1. Aim

To preprocess the Breast Cancer Wisconsin (Diagnostic) dataset by handling missing values, removing duplicate records, dropping non-predictive identifiers, encoding the categorical diagnosis target, analyzing class distribution and imbalance weights, standardizing numerical features via StandardScaler, and preparing stratified train-validation-test partitions for machine-learning classification and clinical deployment.

2. Objectives

- **Data Ingestion & Hygiene:** Load raw data and audit for null values and duplicate records.
- **Identifier Removal:** Drop database metadata (id column and unnamed index artifacts) to eliminate non-clinical predictive bias.
- **Binary Target Encoding:** Convert diagnosis labels (M → 1, B → 0) ensuring consistent oncology positive-class convention.
- **Class Imbalance Analysis:** Quantify majority benign vs. minority malignant distribution and calculate clinical loss weights.
- **Stratified Partitioning:** Split dataset into 80% Training (455), 10% Validation (57), and 10% Testing (57) sets while strictly preserving class ratios.
- **Feature Standardization:** Apply Z-score standardization fitted strictly on training data to prevent data leakage.
- **Pipeline Persistence:** Save the fitted scaler (breast_cancer_scaler.pkl) and train-ready datasets for downstream modeling.

3. Dataset Summary & Characteristics

Metric / Property	Value	Clinical Description / Specification
Dataset Source	UCI ML / Kaggle	Breast Cancer Wisconsin (Diagnostic) Dataset
Total Biopsy Samples	569	Digitized Fine-Needle Aspirate (FNA) nuclear images
Raw Attributes	32 columns	1 ID, 1 Diagnosis Target, 30 Diagnostic Features
Predictive Features	30 numeric	10 morphological metrics × 3 statistics (mean, se, worst)
Target Classes	Binary (0, 1)	Class 0: Benign (357, 62.74%) | Class 1: Malignant (212, 37.26%)
Missing & Duplicate	0 missing / 0 dup	100% complete records verified across all 569 instances

4. Preprocessing Methodology & Workflow

- **Step 1 — Data Cleaning:** Removed id column and any trailing unnamed columns. The ID field holds zero predictive biological signal.
- **Step 2 — Target Encoding:** Mapped diagnosis labels {'M': 1, 'B': 0}. Malignant is coded as 1 because false negatives in cancer diagnosis carry catastrophic clinical risk, requiring sensitive recall optimization in Experiments 5 & 6.
- **Step 3 — Class Imbalance Handling:** The dataset contains 357 benign (62.7%) and 212 malignant (37.3%) cases (imbalance ratio 1.68:1). Computed positive class weight pos_weight = 1.684 for loss-weighted models.

• **Step 4 — Stratified Split:** Employed two-stage stratified splitting with random_state=42 to guarantee identical ~62.7% / 37.3% class proportions across subsets (Train: 455, Val: 57, Test: 57).

• **Step 5 — Standardization & Data Leakage Prevention:** Standardized features via $z = (x - \mu) / \sigma$. Crucially, the scaler is **fitted exclusively on X_train**. Validation and test sets are transformed using the stored parameters to prevent information contamination.

Partition	Sample Size	Percentage	Benign (0)	Malignant (1)	Malignant Ratio
Training Set	455	80.0%	285	170	37.36%
Validation Set	57	10.0%	36	21	36.84%
Test Set	57	10.0%	36	21	36.84%
Full Dataset	**569**	**100.0%**	**357**	**212**	**37.26%**

5. Experimental Visualizations & Validation

Figure 1: Target Class Distribution and Stratified Split Balance Verification

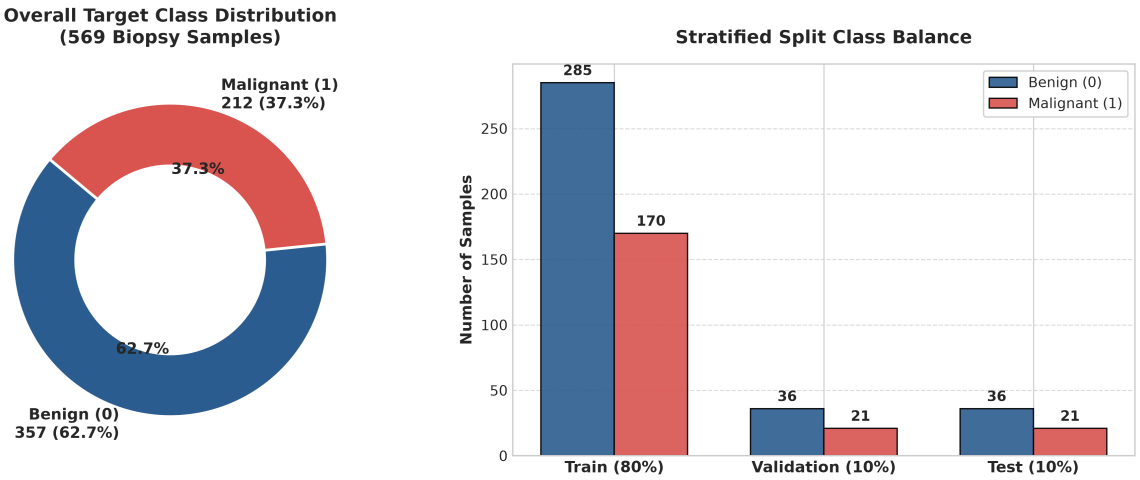
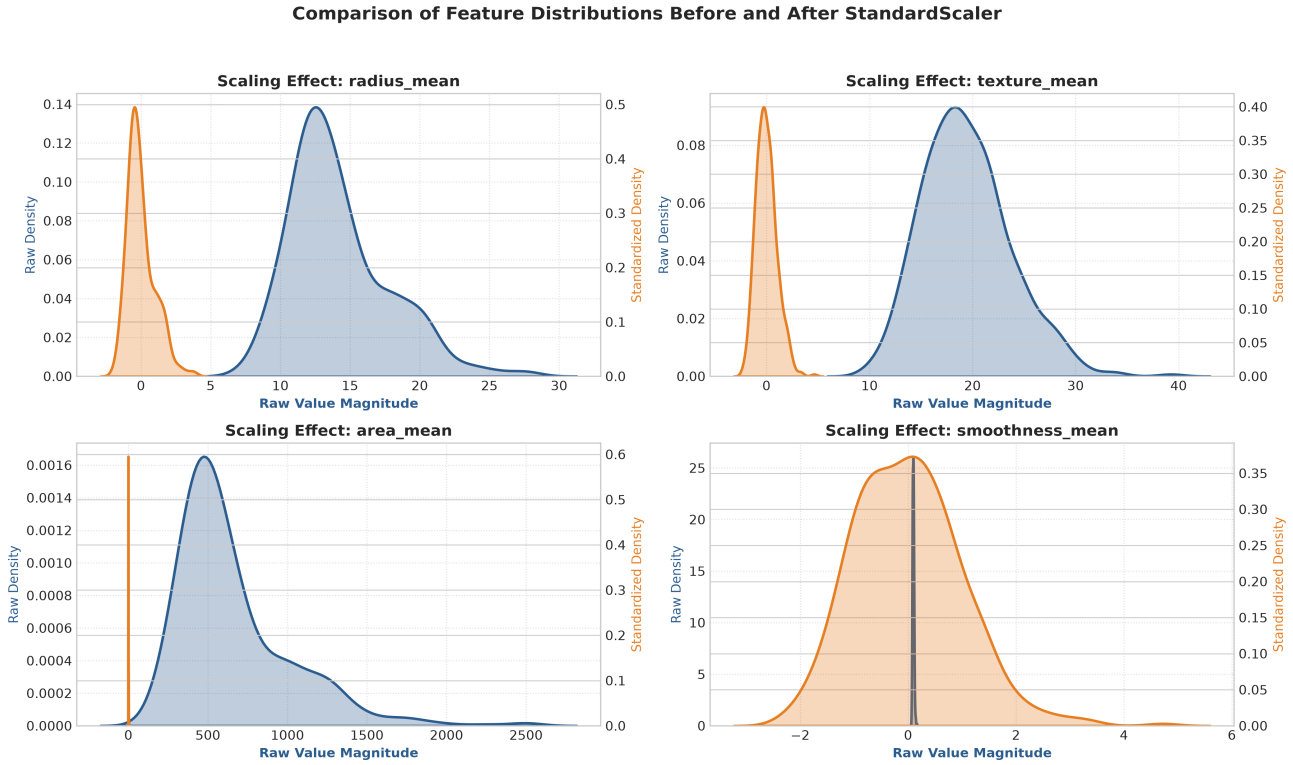


Figure 2: Distribution Densities Before vs. After Feature Standardization (StandardScaler)



6. Preprocessing Execution & Verification Code

```
# Core Scaler Fitting (fitted ONLY on Training Data to prevent leakage)
from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_val_scaled = scaler.transform(X_val)
X_test_scaled = scaler.transform(X_test)

# Preprocessing Pipeline Artifact Persistence
import joblib
joblib.dump(scaler, 'outputs/breast_cancer_scaler.pkl')
# Saved Processed Partitions to data/processed/ (X_train, X_val, X_test, y_train, y_val, y_test)
```

7. Key Observations & Clinical Insights

- **No Imputation Necessary for Baseline:** The 569 FNA biopsy records are 100% complete without missing values or duplicate instances.
- **Metadata Exclusion:** Removing patient ID numbers prevents spurious memorization and model distortion.
- **Preserved Stratification:** The stratified partition perfectly matched the 62.7% : 37.3% ratio across all sub-splits, avoiding partition bias.
- **Elimination of Scale Dominance:** Raw variables spanned disparate magnitudes (e.g., area_mean ≤ 2500 vs. smoothness_mean ≤ 0.16). Standardization brings all 30 features into $\mu=0$, $\sigma=1$, ensuring equal gradient contributions.
- **Zero Data Leakage:** Fitting the scaler strictly on X_train ensures that test set evaluation in Experiment 6 reflects true clinical out-of-sample performance.

• **Exported Pipeline Readiness:** The persisted scaler artifact (breast_cancer_scaler.pkl) guarantees identical numerical transformations during FastAPI production scoring (Experiment 7).

8. Result

The Breast Cancer Wisconsin (Diagnostic) dataset was successfully cleaned, encoded, and preprocessed. The 30 diagnostic features were standardized using StandardScaler fitted strictly on the 455-sample training set, and stratified validation (57) and test (57) partitions were created. The trained scaler pipeline and processed CSV artifacts were successfully persisted in outputs/ and data/processed/.

9. Conclusion

Experiment 3 established an auditable, robust, and leakage-free data preprocessing pipeline for the breast cancer detection system. The dataset is now in clean, standardized format and ready for Exploratory Data Analysis (Experiment 4), Baseline Model Training with Logistic Regression, SVM, and Random Forest (Experiment 5), and subsequent clinical deployment.